

In this Standard the comma has been adopted as the decimal sign

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RIVETS AND PINS - DRILLING, COUNTERSINKING AND DIMPLING

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1. SCOPE

This Standard establishes the dimensional requirements (including tolerances) for drilling, countersinking, and dimpling for metallic parts. See NE 40-273 for composite parts.

2. APPLICATION

This Standard is applicable to the following types of fastener:

- a) Solid rivet;
- b) Blind fastener;
- c) Hi-Lok pin;
- d) Hi-Tigue pin;
- e) Hi-Lite pin;
- f) Lockbolt pin;
- g) Briles rivet;
- h) Pull-Stem and Pull-In pin; and
- i) Radial expansion pin.

3. APPLICABLE DOCUMENTS

Embraer Standards

DSM - Chapter 4 - Mechanical Drawings.

NE 03-085 - Surface Roughness - Inspection.

NE 06-004 - "Hi-Lok" Pins - Requirements for Grip Selection and Installation.

NE 06-012 - "Hi-Tigue" Pins - Requirements for Grip Selection and Installation.

NE 06-013 - Rivet, Solid, 120° Flush Interference Head - Requirements for Grip Selection and Installation.

NE 06-023 - "Lockbolts" Pins - Requirements for Grip Selection and Installation.

NE 06-024 - "Hi-Lite" Pins - Requirements for Grip Selection and Installation.

NE 06-054 - "Pull-Stem/Pull-In" Fastener System Requirements for Grip Selection and Installation.

NE 07-038 - EMB-145/EMB-314/EMB-390/EMB-500/505/EMB-550 and ERJ 170/190 Visual Defects - Rework and Acceptance Criteria for External Areas.

NE 40-006 - Aluminum and Its Alloys - Chemical Conversion.

NE 40-134 - Sealing of Aircraft - General Requirements.

NE 40-135 - Corrosion Resistant Primer for Integral Fuel Tanks (AMS-C-27725).

NE 40-143 - Epoxy Primer, Fluid Resistant for Internal Metallic and Composite Parts - Application Requirements.

NE 40-264 - One-Up Assembly for Robot Applications - Qualification Process Requirements.

NE 40-297 - Sealing - Faying surfaces and fasteners assembly.

NE 42-027 - Parts - Temporary Protection.

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Foreign Standards

American Society for Testing and Materials (ASTM)

ASTM B265 - Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate.

Society of Automotive Engineers (SAE)

AMS 4902 - Titanium Sheet, Strip, and Plate Commercially - Pure Annealed 40.0 ksi (276 MPa) Yield Strength - UNS R50400.

4. DEFINITIONS

4.1 EDGE DISTANCE

Distance between the sheet edge and the center(s) of the hole(s) closer.

4.2 PITCH

Distance between the center of two fasteners immediately closer.

4.3 SCRATCH

Indication of line appearance, continuous or not. Its width is many times smaller if compared to its length and its depth is greater than 10 µm.

4.4 SURFACE DAMAGE

Any indication or mark that may impair the structural integrity of the part.

5. REQUIREMENTS

5.1 PITCH AND EDGE DISTANCE

The nominal pitch and edge distance values shall be as specified on production drawings. See DSM Chapter 04.

The general tolerances for the distances between the center of a hole, used as a datum, and the centers of the other holes shall be as specified in Figure 1.

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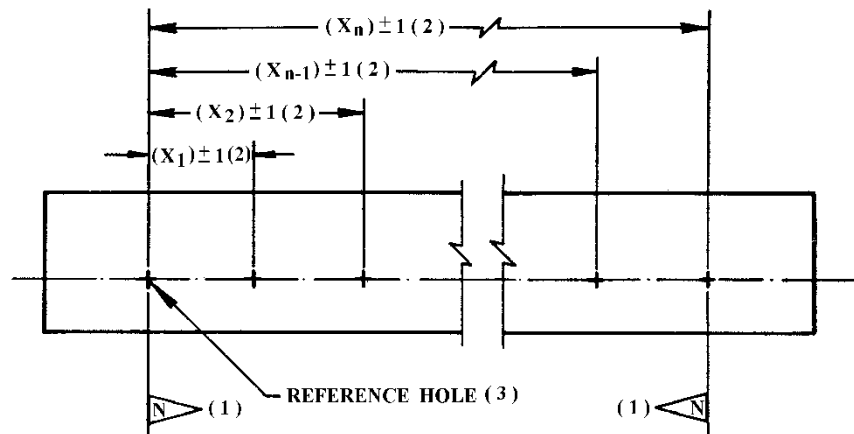
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(1) Indicate specific tolerance, whenever general tolerance (± 1 mm) does not meet design requirements.

(2) The letter “X” means the distance between the reference hole center and the center of the other holes. These dimensions are shown here for illustration purposes only. They are not shown in drawings.

(3) Use the first hole as datum, unless otherwise indicated on drawings.

FIGURE 1 : TOLERANCES BETWEEN HOLE CENTERS

Edge distance tolerances shall be as specified in Figure 2.

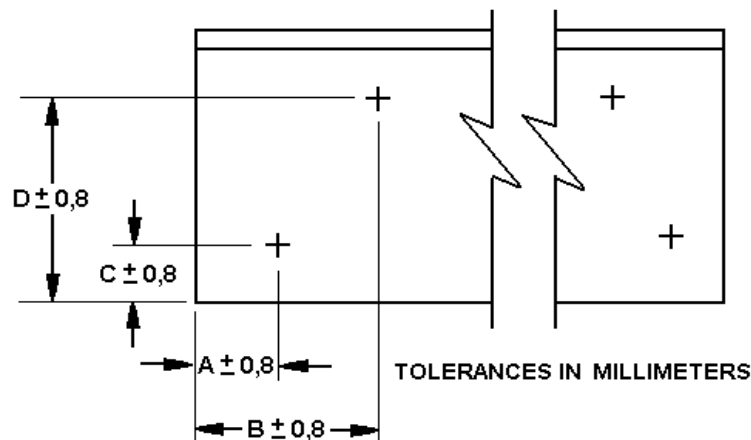
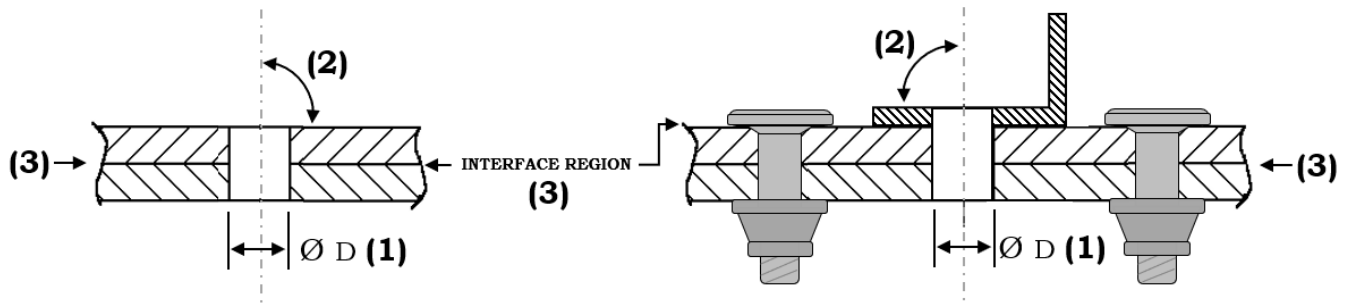


FIGURE 2 : EDGE DISTANCE TOLERANCES

5.2 DRILLING

Figure 3 shows the geometric configuration of the holes recommended for riveting.



(1) To determine hole diameter “ØD”, refer to 5.2.1 through 5.2.7 Centering-holes (guide-holes) dimensioned with Ø2,1 mm on drawings may be drilled (preferably) with Ø2,5 mm, since the manufacturing cost justifies the alteration.

(2) Unless otherwise specified, the centerline of the holes shall be perpendicular to the parts within $\pm 1^\circ$.

(3) Unless otherwise specified, no Gap shall be allowed between the parts.

FIGURE 3 : GEOMETRIC CONFIGURATION OF RIVET HOLE

Ensure that no burrs or debris are present in the holes and on adjacent surfaces prior to fastener and rivet installation.

To eliminate burrs due to the drilling operation, the edges of the holes on each joint component sheet, in the interface region, shall be deburred by sanding with a 220 Aluminum Oxide Cloth. For joints where the installation of rivets has already been performed, only the external surfaces shall be considered as the interface region when a new drilling operation is performed.

This operation shall be as localized as possible so as to avoid the removal of material and/or protection of the sheet in the adjacent regions to the holes. In order to avoid the “knife effect”, the use of drill bit is not allowed.

NOTES

1) The drill bit, files or cutting tools shall be used only when the procedures are not practical due to manufacturing difficulties (e.g. in cases where burrs are not easily removed with aluminum oxide sandpaper) and shall be approved by Manufacturing Engineering.

2) If drilling shall be made on parts with final protection, the holes shall be protected as follows:

a) aluminum and its alloys parts (except polished skin quality): chemical conversion according to NE 40-006, or primer application according to NE 40-143 or NE 40-135 as applicable. If the fastener/hardware is wet assembled using interface sealant according to NE 40-297 within 72 hours after drilling, protection with chemical conversion according to NE 40-006 or primer according to NE 40-143 or NE 40-135 is not necessary;

b) carbon steel or low alloy steel parts: temporary protection according to NE 42-027, unless otherwise specified on drawing; and

c) titanium, stainless steel, or composite materials parts: protection is not necessary.

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Unless otherwise specified, the fastener installation shall be performed according to 40-297 or NE 40-134, as applicable.

NE

-ATTENTION-

For briles type rivet installation (MS14218/PE21118 and MS14219/PE21119), no final surface protection shall be applied in the hole.

5.2.1 Solid Rivets

The dimensional limits for the diameter ØD (Figure 3) of holes shall be according to Table 1.

NOTE

See item 5.2.6 for rivet solid, 120° flush interference head.

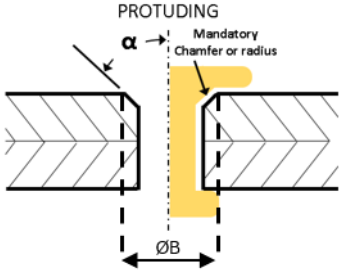
TABLE 1 : DIMENSIONAL LIMITS FOR SOLID RIVET HOLE DIAMETER

RIVET NOMINAL DIAMETER		ØD		
		MS20426, MS20427 MS20470, MS20613 MS20615, NAS1097 NAS1200	CSR 922, CSR 924 CSR 925	CSR 90433
mm	in	mm		
1,6	1/16	1,70 - 1,80	-	-
2,4	3/32	2,49 - 2,69	-	-
3,2	1/8	3,26 - 3,40	-	3,28 - 3,35
4,0	5/32	4,09 - 4,22	4,06 - 4,14	4,04 - 4,12
4,8	3/16	4,88 - 5,03	4,71 - 4,79	4,85 - 4,95
5,6	7/32	5,68 - 5,84	5,60 - 5,67	-
6,4	1/4	6,48 - 6,66	6,24 - 6,31	-
7,9	5/16	8,05 - 8,18	7,82 - 7,90	-
9,5	3/8	9,65 - 9,78	9,41 - 9,49	-

NOTE

For CSR925 installation into titanium or sheet of equivalent hardness, mandatory chamfer or radius is required.

TABLE 2 : DIMENSIONAL LIMITS FOR CHAMFER OR RADIUS OF THE HOLE FOR CSR925 RIVET INSTALLATION INTO TITANIUM OR SHEET OF EQUIVALENT HARDNESS

	RIVET NOMINAL DIAMETER		ØB	α	R
	mm	in	mm		mm
	4,0	5/32	5,21 - 5,46	45°	0,51 - 0,76
	4,8	3/16	6,30 - 6,55		0,64 - 0,89
	6,4	1/4	7,82 - 8,08		0,64 - 0,89
	7,9	5/16	9,65 - 9,91		0,76 - 1,02
	9,5	3/8	11,28 - 11,53		0,76 - 1,02

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5.2.2 Blind Rivets

The dimensional limits for diameter ØD (Figure 3) of the holes shall be according to Table 3.

NOTE

Hole geometry shall ensure the proper seating of the fastener head in accordance with the applicable installation Standard. If necessary, a relief chamfer, equivalent in size up to 50% of the maximum head to shank fillet radius of the fastener may be added in order to accomplish the proper fastener head seating.

TABLE 3 : DIMENSIONAL LIMITS FOR BLIND FASTENER HOLE DIAMETER

RIVET NOMINAL DIAMETER		MS20600, MS20601, NAS1398, NAS1399, M7885/2, M7885/3, M7885/4, M7885/5, CCR244, CCR264, CCR274, NAS9301, NAS9302, NAS9303, NAS9307, NAS9308, NAS9309, NAS1919, NAS1921	NAS1738, NAS1739, NAS1768, NAS1769	BLIND BOLTS: PLT110, PLT210, PLT220, PLT5410, MBF3004, MBF3006, NAS1669, NAS1670, NAS1671, MBF2313	CR7774S CR7684S UAB130-EU CR7683S UABP-EU MS90354 MS90355 CR7771S	MRL3210, MRL3212
mm	in	mm				
2,4	3/32	2,46 - 2,57	-	-	-	-
3,2	1/8	3,28 - 3,35	3,63 - 3,71	-	3,28 - 3,35	-
4,0	5/32	4,06 - 4,17	4,47 - 4,57	4,19 - 4,27	4,17 - 4,24	-
4,8	3/16	4,88 - 4,98	5,21 - 5,31	5,05 - 5,13	5,05 - 5,13	5,080 - 5,131
5,5	7/32	-	-	-	-	5,817 - 5,867
6,4	1/4	6,50 - 6,63	-	6,60 - 6,68	6,60 - 6,68	6,629 - 6,680
7,1	9/32	-	-	-	-	7,391 - 7,442
7,9	5/16	-	-	7,92 - 8,00	-	7,950 - 8,001
8,7	11/32	-	-	-	-	8,763 - 8,814
9,5	3/8	-	-	9,52 - 9,60	-	9,550 - 9,601

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5.2.3 HI-LOK, HI-LITE AND LOCKBOLT PINS

The dimensional limits for the diameter ØD (Figure 3) of holes shall be according to Table 4 in order to allow the best fit class for each class.

TABLE 4 : DIMENSIONAL LIMITS OF HOLE DIAMETER FOR HI-LOK, HI-LITE AND LOCKBOLT LGPL PINS

Ø	ORIGINAL					1 st OVER				2 nd OVER			
	(in) (mm)	PIN		CLASS	HOLE		PIN		HOLE (mm)	PIN		HOLE	
		(in)	(mm)		(mm)		(in)	(mm)		(in)	(mm)	(mm)	
5	5/32" 4,0	0,1625 0,1635	4,128 4,1529	I	4,166	4,267							
				II	4,153	4,178							
				IIA	4,153	4,203							
				III	4,089	4,166							
				IV	4,102	4,128							
				V	4,094	4,110							
				VI	4,078	4,128							
6	3/16" 4,8	0,1885 0,1895	4,788 4,813	I	4,826	4,928	0,2016 0,2026	5,121 5,146	5,159 5,146	0,2172 0,2182	5,517 5,542	5,555 5,542	5,657 5,568
				II	4,813	4,839			5,196			5,542	5,592
				IIA	4,813	4,863			5,159			5,479	5,555
				III	4,750	4,826			5,121			5,491	5,517
				IV	4,762	4,788			5,103			5,484	5,499
				V	4,755	4,770			5,121			5,467	5,517
				VI	4,738	4,788							
8	1/4" 6,4	0,2485 0,2495	6,312 6,337	I	6,350	6,452	0,2641 0,2651	6,708 6,734	6,746 6,733	0,2797 0,2807	7,104 7,130	7,142 7,129	7,244 7,155
				II	6,337	6,363			6,783			7,129	7,179
				IIA	6,337	6,387			6,746			7,066	7,142
				III	6,274	6,350			6,708			7,079	7,104
				IV	6,287	6,312			6,690			7,071	7,086
				V	6,279	6,294			6,708			7,054	7,104
				VI	6,262	6,312							
10	5/16" 7,9	0,3110 0,3120	7,899 7,925	I	7,938	8,039	0,3266 0,3276	8,296 8,321	8,334 8,321	0,3422 0,3432	8,692 8,717	8,730 8,717	8,831 8,742
				II	7,925	7,950			8,371			8,717	8,767
				IIA	7,925	7,975			8,333			8,641	8,729
				III	7,849	7,937			8,295			8,666	8,691
				IV	7,874	7,899			8,278			8,658	8,674
				V	7,866	7,882			8,295			8,641	8,691
				VI	7,849	7,899							
12	3/8" 9,5	0,3735 0,3745	9,487 9,512	I	9,525	9,627	0,3891 0,3901	9,883 9,909	9,921 9,908	0,4047 0,4057	10,279 10,305	10,317 10,304	10,419 10,330
				II	9,512	9,538			9,958			10,304	10,354
				IIA	9,512	9,562			9,921			10,215	10,317
				III	9,423	9,525			9,883			10,254	10,279
				IV	9,462	9,487			9,865			10,246	10,261
				V	9,454	9,469			9,883			10,229	10,279
				VI	9,437	9,487							
14	7/16" 11,1	0,4360 0,4370	11,074 11,100	I	11,112	11,215	0,4516 0,4526	11,471 11,496	11,508 11,495	0,4672 0,4682	11,867 11,892	11,904 11,891	12,007 11,918
				II	11,099	11,126			11,546			11,892	11,942
				IIA	11,100	11,150			11,509			11,802	11,905
				III	11,010	11,113			11,471			11,841	11,867
				IV	11,049	11,075			11,453			11,833	11,849
				V	11,041	11,057			11,470			11,816	11,866
				VI	11,024	11,074							
16	1/2" 12,7	0,4985 0,4995	12,662 12,687	I	12,700	12,802	0,5141 0,5151	13,058 13,084	13,096 13,083	0,5297 0,5307	13,454 13,480	13,492 13,479	13,594 13,505
				II	12,687	12,713			13,133			13,479	13,529
				IIA	12,687	12,737			13,096			13,390	13,492
				III	12,598	12,700			13,058			13,429	13,454
				IV	12,637	12,662			13,040			13,421	13,436
				V	12,629	12,644			13,058			13,404	13,454
				VI	12,612	12,662							
18	9/16" 14,3	0,5605 0,5615	14,237 14,262	I	14,275	14,377	0,5761 0,5771	14,633 14,658	14,671 14,658	0,5917 0,5927	15,029 15,055	15,067 15,054	15,169 15,080
				II	14,262	14,288			14,708			15,054	15,104
				IIA	14,262	14,312			14,671			14,965	15,067
				III	14,173	14,275			14,633			15,004	15,029
				IV	14,212	14,237			14,615			14,996	15,011
				V	14,204	14,219			14,633			14,979	15,029
				VI	14,187	14,237							

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Continued

TABLE 5 : DIMENSIONAL LIMITS OF HOLE DIAMETER FOR HI-LOK, HI-LITE AND LOCKBOLT LGPL PINS

ø	ORIGINAL						1 st OVER				2 nd OVER			
	(in) (mm)	PIN		CLASS	HOLE		PIN		HOLE		PIN		HOLE	
		(in)	(mm)		(mm)		(in)	(mm)	(mm)		(in)	(mm)	(mm)	
2 0	5/8" 15,9	0,6230 0,6240	15,824 15,850	I	15,862	15,965	0,6386 0,6396	16,220 16,246	16,258	16,361	0,6542 0,6552	16,617 16,642	16,654	16,757
				II	15,849	15,876			16,245	16,272			16,641	16,668
				IIA	15,850	15,900			16,246	16,296			16,642	16,692
				III	15,760	15,863			16,156	16,259			16,552	16,655
				IV	15,799	15,825			16,195	16,221			16,591	16,617
				V	15,791	15,807			16,187	16,203			16,583	16,599
				VI	15,774	15,824			16,170	16,220			16,566	16,616
2 4	3/4" 19,1	0,7480 0,7490	18,999 19,025	I	19,037	19,140	0,7636 0,7646	19,395 19,421	19,433	19,536	0,7792 0,7802	19,792 19,817	19,829	19,932
				II	19,024	19,051			19,420	19,477			19,816	19,843
				IIA	19,025	19,075			19,421	19,471			19,817	19,867
				III	18,935	19,038			19,331	19,434			19,727	19,830
				IV	18,974	19,000			19,370	19,396			19,766	19,792
				V	18,966	18,982			19,362	19,378			19,758	19,774
				VI	18,949	18,999			19,345	19,395			19,741	19,791

NOTES

- 1) Perpendicularity between the rivet hole centerline and the sheet: $\pm 0,5^\circ$
- 2) The hole shall have a chamfer specified by NE 06-004, NE 06-023 or NE 06-024, according to the pin type.
- 3) The 5/32" pin first oversize is the 3/16" pin nominal size.
- 4) The 5/32" pin second oversize is the 3/16" pin first oversize.

5.2.3.1 Maximum Grip Length

Table 6 and Table 7 are applicable to interference fit class VI that shall be used for Al-Al joints, only.

The maximum grip length for Hi-Lok, Hi-Lite or Lockbolt LGPL pins installed by pneumatic hammer, shall be two times the pin diameter according to Table 6.

TABLE 6 : MAXIMUM GRIP LENGTH FOR HI-LOK, HI-LITE OR LOCKBOLT LGPL PINS INSTALLED BY PNEUMATIC HAMMER

PIN DIAMETER (IN)	MAXIMUM GRIP LENGTH (DASH)
3/16	-6
1/4	-8
5/16	-10
3/8	-12

NOTE

For other pin diameters, consult Materials Engineering. The maximum grip length for pull type Lockbolt LGPL pins shall be according to Table 7.

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TABLE 7 : MAXIMUM GRIP LENGTH FOR PULL TYPE LOCKBOLT LGPL PINS

PIN DIAMETER (IN)	MAXIMUM GRIP LENGTH (DASH)
3/16	-8
1/4	-11
5/16	-14
3/8	-16

NOTE

For other pin diameters, consult Materials Engineering.

5.2.4 Pull-Stem and Pull-In Pins

The dimensional limits for the diameter ØD (see Figure 3) of holes shall be according to Table 8.

TABLE 8 : DIMENSIONAL LIMITS OF HOLE DIAMETER FOR HSL/HPL™ PINS

	ø	(in) (mm)	HOLE DIAMETER								INTERFERENCE				
			NOMINAL		1 st OVER		2 nd OVER		3 rd OVER				Min	Max	Mean
			(in)	(mm)	(in)	(mm)	(in)	(mm)	(in)	(mm)	(in)	(mm)	(%)	(%)	(%)
PULL STEM PINS	6	3/16" 4,8	0,1865 0,1881	4,738 4,778	0,1996 0,2012	5,071 5,111					0,0004 0,0030	0,010 0,075	0,21	1,59	0,90
	8	1/4" 6,4	0,2463 0,2479	6,257 6,297	0,2620 0,2635	6,654 6,694	0,2776 0,2791	7,050 7,089			0,0006 0,0032	0,015 0,080	0,24	1,28	0,76
	10	5/16" 7,9	0,3088 0,3104	7,844 7,884	0,3244 0,3260	8,240 8,280	0,3400 0,3416	8,636 8,676			0,0006 0,0032	0,015 0,081	0,20	1,03	0,61
	12	3/8" 9,5	0,3711 0,3727	9,427 9,467	0,3868 0,3883	9,824 9,863	0,4022 0,4039	10,217 10,259			0,0008 0,0034	0,020 0,085	0,21	0,90	0,56
	14	7/16" 11,1	0,4333 0,4350	11,007 11,049	0,4489 0,4506	11,403 11,446	0,4645 0,4662	11,799 11,842			0,0010 0,0037	0,025 0,093	0,23	0,84	0,54
	16	1/2" 12,7	0,4958 0,4975	12,594 12,637	0,5115 0,5131	12,991 13,033	0,5270 0,5287	13,387 13,429			0,0010 0,0037	0,025 0,093	0,20	0,74	0,47
PULL-IN PINS	18	9/16" 14,3	0,5576 0,5593	14,164 14,207	0,5732 0,5749	14,560 14,603	0,5889 0,5905	14,957 14,999			0,0012 0,0039	0,030 0,098	0,21	0,69	0,45
	20	5/8" 15,9	0,6202 0,6218	15,752 15,794	0,6357 0,6374	16,148 16,190	0,6513 0,6530	16,544 16,587	0,6826 0,6843	17,339 17,382	0,0012 0,0038	0,030 0,098	0,19	0,62	0,41
	24	3/4" 19,1	0,7446 0,7466	18,912 18,964	0,7602 0,7622	19,309 19,361	0,7758 0,7778	19,705 19,757	0,8071 0,8091	20,500 20,552	0,0014 0,0044	0,035 0,113	0,19	0,60	0,39
	28	7/8" 22,2	0,8694 0,8714	22,083 22,134	0,8850 0,8870	22,480 22,531	0,9006 0,9026	22,876 22,927	0,9319 0,9339	23,671 23,722	0,0016 0,0046	0,040 0,117	0,18	0,53	0,35
	32	1" 25,4	0,9942 0,9962	25,253 25,304	1,0098 1,0119	25,650 25,701	1,0254 1,0274	26,046 26,097	1,0567 1,0587	26,841 26,892	0,0018 0,0048	0,045 0,122	0,18	0,48	0,33

NOTES

- 1) Perpendicularity between pin hole centerline and the structure plane: $\pm 0,5^\circ$.
- 2) Chamfer the holes according to NE 06-054.
- 3) Maximum grip length according to NE 06-054.
- 4) HST112 and HST412 may be used as first and second oversize pins. In this case, use class IV for dimensional holes, according to Table 4.

5.2.5 Hi-Tigue Pins

The dimensional limits for the diameter ØD (Figure 3) of holes shall be according to Table 9.

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**TABLE 9 : DIMENSIONAL LIMITS OF HI-TIGUE PIN HOLE DIAMETER
(NOT APPLICABLE TO HI-TIGUE R.I.F. PINS)**

PIN NOMINAL DIAMETER		PIN DIAMETER DIMENSIONAL LIMITS	ØD INTERFERENCE FIT
mm	in	mm	
4,0	5/32	4,280 - 4,305	4,204 - 4,229
4,8	3/16	4,940 - 4,966	4,864 - 4,890
6,4	1/4	6,464 - 6,490	6,388 - 6,414
7,9	5/16	8,052 - 8,077	7,976 - 8,001
9,5	3/8	9,639 - 9,665	9,563 - 9,589

NOTES

- 1) Perpendicularity between the pin hole centerline and the sheet: $\pm 0,5^\circ$.
- 2) The hole shall have a chamfer specified by NE 06-012.

5.2.6 Rivet Solid, 120° Flush Interference Head

The dimensional limits for the diameter ØD (Figure 3) of holes shall be according to Table 10.

TABLE 10 : DIMENSIONAL LIMITS FOR HOLE DIAMETER OF RIVET, SOLID, 120°, FLUSH INTERFERENCE HEAD (MS14218, MS14219, PE 2.6.118 AND PE 2.6.119)

RIVET NOMINAL DIAMETER		ØD
mm	in	mm
2,4	3/32	2,49 - 2,64
3,2	1/8	3,25 - 3,43
4,0	5/32	4,09 - 4,24
4,8	3/16	4,90 - 5,08
5,6	7/32	5,79 - 5,97
6,4	1/4	6,53 - 6,73
7,1	9/32	7,37 - 7,57
7,9	5/16	8,03 - 8,31

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5.2.7 Radial Expansions Pins

The dimensional limits for the diameter ØD (see Figure 3) of holes shall be according to Table 11.

TABLE 11 : DIMENSIONAL LIMITS FOR HOLE DIAMETER OF RADIAL EXPANSION PINS

Ø	RXL12BP and RXL4BC					ORXB12TP and ORXB4BC				OORXB12TP and OORXB4BC			
	NOMINAL					1 st OVER				2 nd OVER			
	(in) (mm)	SLEEVE		HOLE		SLEEVE		HOLE		SLEEVE		HOLE	
		(in)	(mm)	(in)	(mm)	(in)	(mm)	(in)	(mm)	(in)	(mm)	(in)	(mm)
6	3/16	0,2045	5,194	0,2074	5,268	0,2205	5,601	0,2234	5,674	0,2365	6,007	0,2394	6,081
	4,8	0,2055	5,220	0,2094	5,319	0,2215	5,626	0,2254	5,725	0,2375	6,032	0,2414	6,132
8	1/4"	0,2655	6,744	0,2684	6,817	0,2805	7,125	0,2834	7,198	0,2965	7,531	0,2994	7,605
	6,4	0,2665	6,769	0,2704	6,868	0,2815	7,150	0,2854	7,249	0,2975	7,556	0,3014	7,656
10	5/16"	0,3265	8,293	0,3294	8,367	0,3425	8,699	0,3454	8,773	0,3585	9,106	0,3614	9,180
	7,9	0,3275	8,318	0,3314	8,418	0,3435	8,725	0,3474	8,824	0,3595	9,131	0,3634	9,230
12	3/8"	0,3905	9,919	0,3934	9,992	0,4055	10,300	0,4084	10,373	0,4215	10,706	0,4244	10,780
	9,5	0,3915	9,944	0,3954	10,043	0,4065	10,325	0,4104	10,424	0,4225	10,731	0,4264	10,831
14	7/16"	0,4525	11,493	0,4554	11,567	0,4685	11,900	0,4714	11,974	0,4845	12,306	0,4874	12,380
	11,1	0,4535	11,519	0,4574	11,618	0,4695	11,925	0,4734	12,024	0,4855	12,332	0,4894	12,431
16	1/2"	0,5155	13,094	0,5184	13,167	0,5305	13,475	0,5334	13,548	0,5465	13,881	0,5494	13,955
	12,7	0,5165	13,119	0,5204	13,218	0,5315	13,500	0,5354	13,599	0,5475	13,906	0,5514	14,006

5.2.8 Automation - One Up Assembly Process

All joints that will not be disassembled for cleaning, deburring, or inspection shall have their process qualification according to NE 40-264, unless otherwise specified on drawing.

5.2.9 Scratches and Surface Damage in Holes for Pins

No scratch and surface damage into the hole or at their edges is permitted.

5.2.10 Scratches and Surface Damage on Aluminum Clad Skin Parts.

Scratches and surface marks inherent to drilling process on aluminum clad skin parts that do not pierce the clad layer to expose the core (base metal), should be reworked according to NE 07-038.

NOTE

To check core exposure, the missing clad test according to NE 07-038 may be applied.

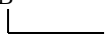
5.3 COUNTERSINKING

Figure 4 shows the countersink geometric configuration for countersunk head rivets.

NOTES

1) Tables 11, 12, 14, 15, 16, 17 and 18 show the values of t_{min} , taking as a reference:

$$t_{min} = 1,5 B$$


 rivet countersunk head height

These values shall not be modified without authorization of Aeronautic Engineering Division.

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2) Countersinking is preferred over dimpling whenever $h = \frac{t}{3}$

3) For joints with both sides countersunk, the countersunk of the driven head shall be equal to that of the manufactured head, unless otherwise specified.

4) If countersinking shall be made on parts with final protection, countersunk shall be protected as follows:

a) aluminum and its alloys parts (except polished skin quality): chemical conversion according to NE 40-006, or primer application according to NE 40-143.

b) If the fastener/hardware is wet assembled using interface sealant according to NE 40-297 within 72 hours after the countersinking, protection with chemical conversion according to NE 40-006 or primer according to NE 40-143 is not necessary.

c) carbon steel or low alloy steel parts: temporary protection according to NE 42-027, unless otherwise specified on drawing; and

d) titanium, stainless steel or composite material parts: protection is not necessary.

Unless otherwise specified, the fastener installation shall be performed according to NE 40-297 or NE 40-134, as applicable.

-ATTENTION-

For briles type rivet installation, no final surface protection shall be applied in the countersunk.

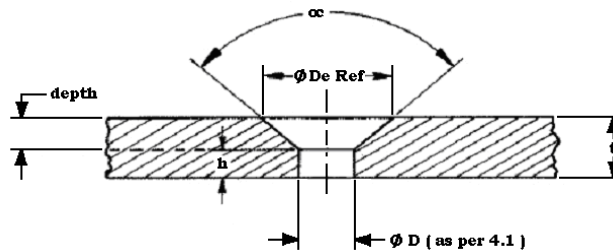


FIGURE 4 : GEOMETRIC CONFIGURATION OF COUNTERSUNK RIVET HOLE

NOTE

ØDe REF according to Tables 11, 12, 14, 15, 17 and 18. The allowable countersunk depth shall be up to 2/3 of local sheet thickness to avoid "knife effect".

5.3.1 Solid Rivets

The dimensional limits for the diameter ØDe, for α angle (Figure 4) and the minimum sheet thickness shall be according to Tables 11, 12 and 14.

NOTE

See item 5.3.4 for briles rivets.

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TABLE 12 : COUNTERSINK DIAMETER AND ANGLE DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FOR MS20426, MS20427 AND CSR922 SOLID RIVETS

NOMINAL RIVET DIAMETER		MS 20426			MS 20427			CSR 922 (2)		
		ØDe REF (1)	t min	α	ØDe REF (1)	t min	α	ØDe REF (1)	t min	α
mm	in	mm			mm			mm		
1,6	1/16	2,5	0,84	100° ±0,5°	3,42	1,21	100° ±1,5°	-	-	100° ±1,0°
2,4	3/32	4,2	1,37		4,83	1,59		-	-	
3,2	1/8	5,4	1,60		5,99	1,84		-	-	
4,0	5/32	6,9	2,10		7,56	2,34		7,3	1,94	
4,8	3/16	8,6	2,67		9,28	2,93		9,0	2,63	
6,4	1/4	11,7	3,62		12,42	3,94		12,1	3,62	
7,9	5/16	14,0	4,04		14,53	4,37		14,3	4,04	
9,5	3/8	17,3	5,11		17,97	5,47		17,6	5,11	

(1) ØDe is a reference dimension. Flushness requirements shall be according to the fastener installation specifications.

(2) For installation into titanium or sheet of equivalent hardness, mandatory chamfer or radius is required according to Table 14.

TABLE 13 : COUNTERSINK DIAMETER AND ANGLE DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FOR NAS1097 AND CSR924 SOLID RIVETS

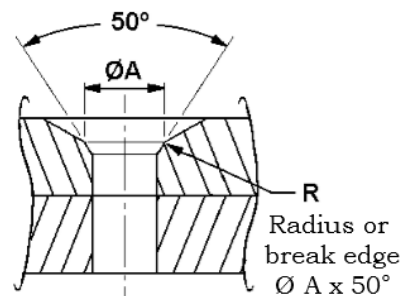
NOMINAL RIVET DIAMETER		NAS1097, NAS1200 (2)		CSR24 (2)			
		ØDe REF (1)	t _{min}	α	ØDe REF (1)	t _{min}	α
mm	in	mm			mm		
2,4	3/32	3,5	0,80	100° ±0,5°	-	-	100° ±1°
3,2	1/8	4,6	1,09		-	-	
4,0	5/32	5,9	1,42		6,2	1,3	
4,8	3/16	7,3	1,75		7,6	1,75	
6,4	1/4	9,7	2,29		10	2,29	
7,9	5/16	-	-		12	2,55	
9,5	3/8	-	-		14,2	2,93	

(1) ØDe is a reference dimension. Flushness requirements shall be according to the fastener installation specifications.

(2) For installation into titanium or sheet of equivalent hardness, mandatory chamfer or radius is required according to Table 14.

TABLE 14 : DIMENSIONAL LIMITS FOR R RADIUS OR FOR BREAKING EDGE FOR RIVETS INSTALLATION

RIVET NOMINAL DIAMETER		ØA	R
mm	in	mm	
4,0	5/32	4,70 - 4,95	0,64 - 0,89
4,8	3/16	5,46 - 5,72	0,76 - 1,02
6,4	1/4	6,98 - 7,24	0,76 - 1,02
7,9	5/16	8,76 - 9,02	1,02 - 1,27
9,5	3/8	10,29 - 10,54	1,02 - 1,27



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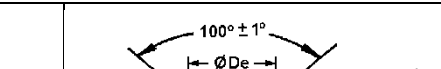
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TABLE 15 : COUNTERSINK DIAMETER DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FORT CSR90433 SOLID RIVET

RIVET NOMINAL DIAMETER		ØDe REF (1)	tmin	
mm	in	mm		
3,2	1/8	4,9	1,07	
4,0	5/32	6,2	1,41	
4,8	3/16	7,6	1,75	
5,6	7/32 (2)	8,2	1,75	

(1) ØDe is a reference dimension. Flushness requirements shall be according to the fastener installation specifications.

(2) Ø7/32 in rivet is used as replacement repair of Ø3/16 in rivet.

5.3.2 Blind Rivets

The dimensional limits for the diameter ØDe and the α angle (see Figure 4), and the minimum sheet thickness shall be according to Table 16.

TABLE 16 : COUNTERSINKING DIAMETER AND ANGLE DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FOR BLIND RIVETS (2)

RIVET NOMINAL DIAMETER		RIVET DIAMETER DIMENSIONAL LIMITS	ØDe REF.	tmin	α
mm	in	mm			
M7885/3, M7885/5, NAS1921, NAS9302, NAS9308					
3,2	1/8	-	5,7	1,60	100° ±0,5°
4,0	5/32	-	7,3	2,10	
4,8	3/16	-	9,0	2,67	
6,4	1/4	-	12,1	3,62	
BLIND BOLTS PLT110, NAS1670 (1)					
4,17	0,164	4,128 - 4,178	8,3	2,63	100° ±1°
5,03	0,198	5,004 - 5,055	9,7	2,93	
6,58	0,259	6,553 - 6,604	12,8	3,89	
BLIND BOLTS: PLT5410					
4,17	0,164	4,128 - 4,178	6,5	1,52	100° ±1°
5,03	0,198	5,004 - 5,055	7,5	1,79	
6,58	0,259	6,553 - 6,604	9,9	2,32	
7,89	0,310	7,861 - 7,912	11,9	2,59	
9,49	0,374	9,462 - 9,512	14,1	2,93	
BLIND BOLTS MBF3006 (1)					
4,0	5/32	4,128 - 4,178	8,3	1,49	130° ±1°
4,8	3/16	4,991 - 5,042	9,7	1,64	
6,4	1/4	6,541 - 6,591	12,8	2,17	

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TABLE 16 : COUNTERSINKING DIAMETER AND ANGLE DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FOR BLIND RIVETS (2)

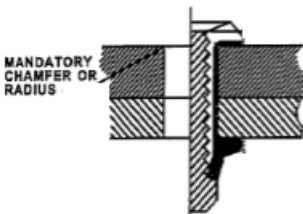
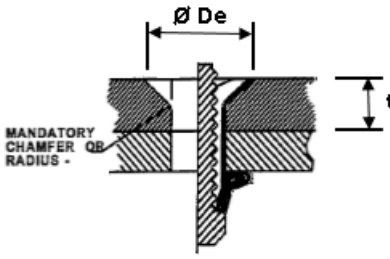
RIVET NOMINAL DIAMETER		RIVET DIAMETER DIMENSIONAL LIMITS	ØDe REF.	tmin	α
mm	in	mm			
BLIND BOLTS: CR7774S, CR7684S, UAB130-EU					
4,0	5/32	4,115 - 4,166	8,5	1,49	130° ±1°
4,8	3/16	5,004 - 5,055	9,8	1,64	
6,4	1/4	6,553 - 6,604	12,9	2,17	
NAS9303 NAS9309					
3,2	1/8	-	4,6	1,09	100° ±0,5°
4,0	5/32	-	5,9	1,42	
4,8	3/16	-	7,3	1,75	
6,4	1/4	-	9,7	2,29	

(1) Included the rivet diameter dimensional limits for PLT110, NAS1670, and MBF3006 Blind Bolts to identify correctly the fastener regarding to its diameter.

(2) For installation into titanium or sheet of equivalent hardness, mandatory chamfer or radius is required according to Table 14.

For MRL3210 and MRL3212 rivets requirements are shown in Table 17.

TABLE 17 : COUNTERSINKING DIAMETER AND MANDATORY CHAMFER LIMITS AND MINIMUM SHEET THICKNESS FOR MRL3210 AND MRL3212 BLIND RIVETS

			
PROTRUDING		CSK HEAD	

RIVET NOMINAL DIAMETER		ØDe REF.	t min	MANDATORY CHAMFER OR RADIUS	
				PROTRUDING	CSK HEAD
mm	in	mm			
4,8	3/16"	9,60	2,93	1,27	1,00
5,5	7/32"	10,39	2,93	1,27	1,00
6,4	1/4"	12,67	3,96	1,40	1,00
7,1	9/32"	13,46	3,96	1,40	1,00
7,9	5/16"	15,90	5,18	1,52	1,27
8,7	11/32"	16,69	5,18	1,52	1,27
9,5	3/8"	19,10	6,17	1,52	1,27

For CCR244 and CCR264 rivets, the minimum depths required to ensure flushness and minimum sheet thickness are shown in Table 18.

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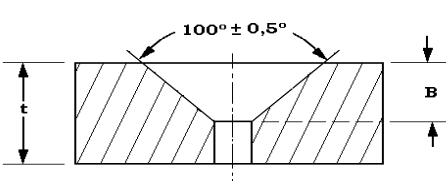
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TABLE 18 : COUNTERSINK DEPTH DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FOR CCR244 AND CCR264 BLIND RIVETS

RIVET TYPE	RIVET NOMINAL DIAMETER		Bmin (1)	tmin	
	mm	in			
CCR244	2,4	3/32	0,53	0,80	
	3,2	1/8	0,71	1,07	
CCR264	2,4	3/32	0,91	1,37	
	3,2	1/8	1,07	1,60	

(1) B is a reference dimension. Flushness requirements shall be according to fasteners installation specification.

5.3.3 Hi-Lok, Hi-Lite, Hi-Tigue and Lockbolt Pins

The dimensional limits for the diameter ØDe and angle α (see Figure 4) and the minimum sheet thickness shall be according to Table 19.

TABLE 19 : COUNTERSUNK DIAMETER AND ANGLE DIMENSIONAL AND MINIMUM SHEET THICKNESS FOR HI-LOK, HI-LITE, HI-TIGUE AND LOCKBOLT LGPL PINS (3)

PIN NOMINAL DIAMETER		ØDe REF. (1)		tmin		α
mm	in	mm				
		HI-LOK / HI-LITE / HI-TIGUE (2)				100° ±0,5°
		SHEAR	TENSION	SHEAR	TENSION	
4,0	5/32	6,8	8,3	1,56	2,67	
4,8	3/16	7,7	9,6	1,79	3,07	
6,4	1/4	10,0	12,8	2,32	4,11	
7,9	5/16	12,0	15,9	2,54	5,14	
9,5	3/8	14,2	19,2	2,97	6,17	
11,1	7/16	16,9	22,5	3,7	7,2	
12,7	1/2	19,1	25,7	4,1	8,2	
		LOCKBOLT FOR METALLIC MATERIALS				
		LGPL2SC-V, LGPS18SCV	LGPL4SC-DT	LGPL2SC-V, LGPS18SCV	LGPL4SC-DT	
4,0	5/32	6,5	7,1	1,60	1,87	
4,8	3/16	7,4	7,9	1,80	2,13	
6,4	1/4	9,8	10,6	2,40	2,83	
7,9	5/16	11,8	13,2	2,54	3,51	
9,5	3/8	14,6	15,8	3,00	4,19	

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TABLE 19 : COUNTERSUNK DIAMETER AND ANGLE DIMENSIONAL AND MINIMUM SHEET THICKNESS FOR HI-LOK, HI-LITE, HI-TIGUE AND LOCKBOLT LGPL PINS (3)

PIN NOMINAL DIAMETER		ØDe REF. (1)		tmin		α
mm	in	mm				
		LOCKBOLT/ HI-LITE FOR COMPOSITE MATERIALS				130° ±0,5°
		HST315, LGPL9SC-V		HST315, LGPL9SC-V		
4,0	5/32	8,5		1,56		
4,8	3/16	9,8		1,79		
6,4	1/4	12,9		2,32		
7,9	5/16	15,2		2,59		
9,5	3/8	17,9		2,97		
11,1	7/16	21,5		3,69		

(1) B is a reference dimension. Flushness requirements shall be according to fastener installation specification.

(2) Except HL23 e HST23. Request for the Aeronautic Engineering Division.

(3) Countersunk bottom shall have a chamfer specified by NE 06-004, NE 06-012, NE 06-023 or NE 06-024, according to the pin type.

5.3.4 Briles Rivets

The dimensional limits for the diameter ØDe and angle α and the minimum sheet thickness shall be according to Table 20.

TABLE 20 : COUNTERSINK DIAMETER DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FOR BRILES BRFZ AND BRFR RIVETS (MS14218, MS14219, PE26118 AND PE26119)

Technical drawing of a countersunk rivet. The drawing shows a cross-section of the rivet head and the sheet metal it is being driven into. Key dimensions labeled are: α (countersink angle), $\varnothing De$ (nominal diameter), r (fillet radius), R (counterbore radius), B (sheet thickness), and t (sheet thickness). A note specifies $r(\max) = 0,25 \text{ mm}$.

RIVET NOMINAL DIAMETER		$\varnothing De$ REF.	B REF (1)	tmin	R $\pm 0,102$	α	
mm	in	mm				120° ±0,5°	
		BRFZ SERIES (MS14218 AND PE26118)					
2,4	3/32	3,137 - 3,213	0,56	0,84	0,203		
3,2	1/8	4,128 - 4,204	0,69	1,03	0,203		
4,0	5/32	5,347 - 5,436	0,89	1,27	0,229		

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TABLE 20 : COUNTERSINK DIAMETER DIMENSIONAL LIMITS AND MINIMUM SHEET THICKNESS FOR BRILES BRFZ AND BRFR RIVETS (MS14218, MS14219, PE26118 AND PE26119)

RIVET NOMINAL DIAMETER		ØDe REF.	B REF (1)	tmin	R ±0,102	α
mm	in	mm				
		BRFZ SERIES (MS14218 AND PE26118)				120° ±0,5°
4,8	3/16	6,566 - 6,668	1,12	1,60	0,305	
5,6	7/32	7,938 - 8,052	1,35	2,02	0,381	
6,4	1/4	9,106 - 9,246	1,55	2,32	0,457	
7,1	9/32	10,300 - 10,439	1,75	2,63	0,508	
7,9	5/16	11,544 - 11,684	1,96	2,93	0,559	
		BRFR SERIES (MS14219 AND PE26119)				120° ±1°
2,4	3/32	4,128 - 4,204	0,86	1,30	0,279	
3,2	1/8	5,347 - 5,436	1,04	1,56	0,356	
4,0	5/32	6,566 - 6,655	1,35	2,02	0,508	
4,8	3/16	7,938 - 8,039	1,73	2,59	0,660	
5,6	7/32	9,284 - 9,385	1,96	2,93	0,838	
6,4	1/4	10,858 - 10,973	2,29	3,43	0,991	
7,1	9/32	12,433 - 12,548	2,54	3,81	1,118	
7,9	5/16	13,246 - 13,360	2,64	3,96	1,270	

(1) Set up the countersinking tool depth, checking flush condition of the gage in accordance with item 6.3. Flushness requirements shall be according to NE 06-013.

5.4 DIMPLING

Dimpling shall be avoided to new designs, but for sheets too thin to countersink, its indication on drawing shall be approved by Materials Engineering. Dimpling shall be performed according to Table 21.

NOTE

Non clad 2024-T3 aluminum alloy shall be dimpled at room temperature.

TABLE 21 : DIMPLING PARAMETERS

SHEET MATERIAL	THICKNESS (in)	PART TEMPERATURE	TOOLING TEMPERATURE	DWELL PRESSURE (psi)	DWELL TIME (s)
ALCLAD 2024-T3	0,025	155 °C ±10 °C	205 °C (MAX)	60	30 to 35
	0,032			70	
	0,040			82	
	0,050			96	55 to 60

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TABLE 21 : DIMPLING PARAMETERS

SHEET MATERIAL	THICKNESS (in)	PART TEMPERATURE	TOOLING TEMPERATURE	DWELL PRESSURE (psi)	DWELL TIME (s)
ALCLAD 2024-T3	0,063	155 °C ±10 °C	205 °C (MAX.)	100	55 to 60
	0,080			110	
ALCLAD 7475-T761	0,025			60	30 to 35
	0,032			70	
	0,040			82	
	0,050			96	55 to 60
	0,063			100	
	0,080			110	
	0,025			40	1 to 5
3XX SERIES CRES ANNEALED	0,032	ROOM TEMPERATURE	ROOM TEMPERATURE	50	
	0,040			60	
	0,050			70	
	0,063			70	
CP TITANIUM (1) 6Al-4V ANNEALED	0,025	270 °C ±10 °C	433 °C ±7 °C	40	2 to 6
	0,032			50	
	0,040			60	
	0,050			70	
	0,063			90	

(1) For commercially pure titanium UNS R 50400 (AMS 4902, and ASTM B265 grade 1, dimpling shall be at room temperature with subsequent visual inspection.

All dimpling for rivets shall consist of the following steps:

- Pilot drilling, according to Table 23;
- Deburring of the downside of pilot hole with countersink (105° ±5°) not exceeding 25% of the total sheet thickness;
- Dimpling according to item 5.4 and Table 21; and
- Drilling to the finished diameter, according to item 5.2.

TABLE 22 : DRILL BIT SIZES FOR DRILLING OF PILOT HOLES AND DRILLING TO THE FINAL HOLE DIAMETER

TYPES OF PINS	PIN NOMINAL DIAMETER		PILOT HOLE DRILL BIT DIAMETER		FINAL HOLE DIAMETER AFTER DIMPLING
	mm	in	No/(in)	mm	
SOLID, BLIND (EXCEPT BLIND BOLTS)	1,6	1/16	51	1,7	According to Table 1 and Table 2
	2,4	3/32	40	2,44	
	3,2	1/8	30	3,26	
	4,0	5/32	21	4,04	
	4,8	3/16	11	4,85	
	6,4	1/4	(1/4)	6,35	
	7,9	5/16	(5/16)	7,94	
	9,5	3/8	(3/8)	9,52	

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TABLE 22 : DRILL BIT SIZES FOR DRILLING OF PILOT HOLES AND DRILLING TO THE FINAL HOLE DIAMETER

TYPES OF PINS	PIN NOMINAL DIAMETER		PILOT HOLE DRILL BIT DIAMETER		FINAL HOLE DIAMETER AFTER DIMPLING
	mm	in	No/(in)	mm	
BLIND BOLTS	4,0	5/32	27	3,66	According to Table 3
	4,8	3/16	24	3,86	
	6,4	1/4	5	5,23	
HI-LOK, HI-LITE LOCKBOLT LGPL AND HI-TIGUE	4,0	5/32	27	3,66	According to Table 4 and Table 8.
	4,8	3/16	24	3,86	
	6,4	1/4	5	5,23	
	7,9	5/16	(1/4)	6,35	
	9,5	3/8	(5/16)	7,94	

Figure 5 shows the geometric configuration of recommended dimpling for flush head pins.

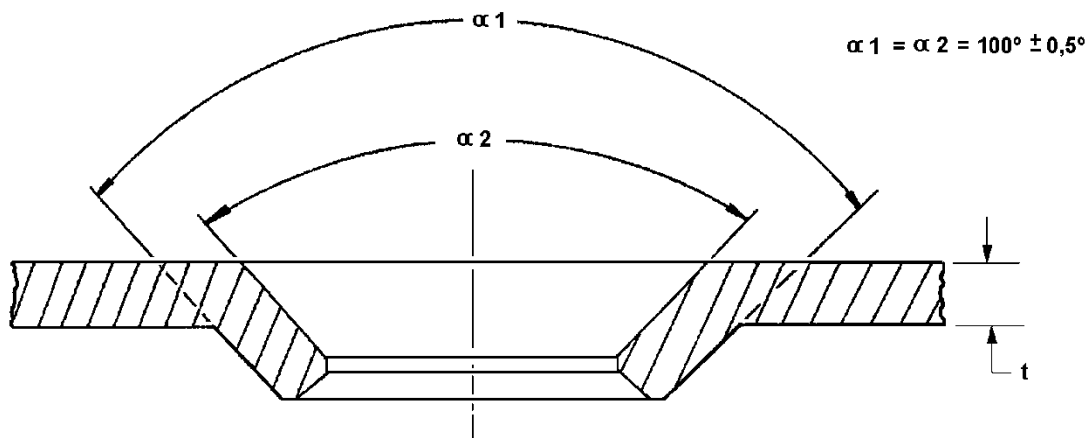


FIGURE 5 : PINS DIMPLING GEOMETRIC CONFIGURATION

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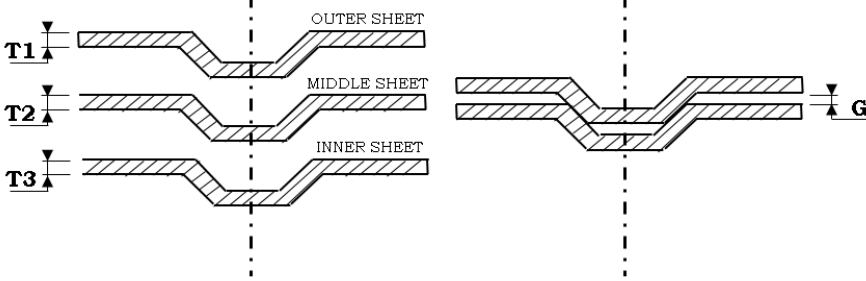
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5.4.1 Multiple Dimpling

The dimensional limits and the other dimensional requirements associated with multiple dimpling for riveting shall be according to Table 23.

TABLE 23 : DIMENSIONAL REQUIREMENTS FOR MULTIPLE DIMPLING

			
RIVET NOMINAL SIZE mm (in)		SHEET THICKNESS MAXIMUM (t ₁ , t ₂ , or t ₃)	GAP G
mm	in	mm	
1,6	1/16	1,02	0,01 - 0,13
2,4	3/32	1,27	0,01 - 0,13
3,2	1/8	1,60	0,01 - 0,25
4,0	5/32	1,80	0,01 - 0,25
4,8	3/16	2,29	0,01 - 0,25
6,4	1/4	3,18	0,01 - 0,41

NOTES

- 1) Pilot hole drilling for sheets dimpling shall be performed according to Table 22.
- 2) Dimpling of extruded parts is not allowed, except where specifically indicated on the engineering drawing. Dimpling dimensions shall be adjusted in the equipment used for dimpling so as to meet the following requirements:
 - a) The angle shall be within tolerances; and
 - b) Manufactured countersunk head height of rivet concerning the sheet, after driving, shall meet the requirements to the fastener installation specifications.
 - c) Rivet head, when protruding above the sheet surface, shall be shaved to meet the requirements specified on fastener installation specifications.
- 3) Dimpled sheets shall be installed with countersunk sheet. See item 5.5.

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5.5 COMBINED COUNTERSINKING AND DIMPLING

Combined countersinking/dimpling is the installation of dimpled and countersunk sheets, and dimensions and tolerances of countersinks for the installation with dimpled sheets shall be according to Table 24.

Dimpling dimensions and gap between sheets shall be according to Table 23.

TABLE 24 : DIMENSIONAL REQUIREMENTS FOR COMBINED COUNTERSINKING AND DIMPLING

The image contains two technical diagrams illustrating rivet installation. The left diagram shows a rivet being installed into a hole with a countersink. The hole diameter is labeled $\varnothing D_e$ and the countersink angle is labeled α . The right diagram shows a rivet being installed into a hole with a countersink. The hole diameter is labeled $\varnothing D$ and the countersink angle is labeled α .

RIVET NOMINAL DIAMETER		HOLE DIAMETER FOR DIMPLING OR COUNTERSINKING	$\varnothing D_e$ REF. (1)	t_{min} for COUNTERSINKING	α
mm	in		mm		
1,6	1/16	ACCORDING TO Table 21 and 5.2	2,97 - 3,23	0,81	100° ±0,5°
2,4	3/32		4,24 - 4,50	1,02	
3,2	1/8		5,46 - 5,72	1,27	
4,0	5/32		7,19 - 7,44	1,6	
4,8	3/16		8,86 - 9,12	2,03	
6,4	1/4		11,96 - 12,22	2,54	
7,9	5/16		14,43 - 14,68	3,18	

(1) ØDe is a reference dimension.

NOTE

Dimpling shall be performed according to the requirements of 5.4.

5.6 SURFACE ROUGHNESS (EXCEPT FOR DIMPLING)

Ra measured along hole axis and countersunk area according to NE 03-085, shall be 1,6 µm maximum.

Rmax shall not be greater than 10 µm. For measures higher than 10 µm, it is considered as a scratch.

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6. TESTS AND INSPECTIONS

6.1 GENERAL

An accurate drilling, countersinking and / or dimpling is a necessary condition to obtain a proper subsequent installation of the rivet.

Whenever possible, it shall be used in drilling / countersinking / dimpling operation and inspection the tool and gages furnished by rivet manufacturers, according to example show in Table 25, for “Briles” type solid rivets.

6.2 HOLE INSPECTION

A visual inspection shall be performed with magnifying lens up to 8X to verify compliance with item 5.2.9.

NOTE

A proven statistical control method shall be applied to visual inspection or other suitable inspection method under previous approval of Embraer Quality and Product Engineering.

6.3 DRILLING AND COUNTERSINKING CHECKING FOR “BRILES” TYPE SOLID RIVETS

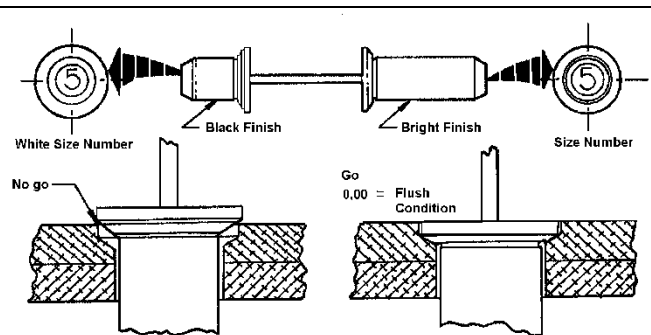
The drilling and countersunk conformity with the requirements herein specified shall be guaranteed by the methods described in 6.3.1 or 6.3.2 or 6.3.3 or any other approved by Quality Engineering.

6.3.1 Go No Go Gage

The drilling and countersunk conformity may be checked in a test specimen with the same thickness of the sheet to be drilled and countersunk, with a “Go-no-Go” gage furnished by the rivet manufacturer. See Table 25.

TABLE 25 : DUAL GAGE (GO-NO-GO) FOR “BRILES” TYPE SOLID RIVET DRILLING AND COUNTERSUNK

RIVET NOMINAL DIAMETER		DUAL GAGE	
		For BRFZ MS14218, PE26118	For BRFR MS14219, PE26119
mm	in		
2,4	3/32	FZDG-3	FRDG-3
3,2	1/8	FZDG-4	FRDG-4
4,0	5/32	FZDG-5	FRDG-5
4,8	3/16	FZDG-6	FRDG-6



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6.3.2 Countersink Depth Gauge

A direct countersink depth measure may be obtained by using a digital countersink depth gauge approved by Manufacture Engineering and Quality Engineering. The utilization of the gauges has the objective to perform the measure of a wide range of countersunk holes in a simple way, and it's only necessary to put the probe into the countersunk and read the depth at the display. The gauge peak and the standard block shall be compatible with the fastener type and size.

The depth gauge shall be used to determine the depth of the countersink relative to “Briles” (of 1/8”, 5/32” and 3/16” nominal diameters) and “Hi-Lite and Lockbolt” (of 5/32”, 3/16”, 1/4”, 5/16” and 3/8” nominal diameters) rivets. Table 25 and Figure 5 inform the allowed deviation for the control of depths of fastener countersinks.

NOTE

The reading peaks and standard blocks shall be identified in order to prevent errors during their utilization as regards the fastener type and diameter.

TABLE 26 : ALLOWED DEVIATION FOR THE CONTROL OF DEPTHS OF FASTENERS COUNTERSINKS

120 °Countersunk				
Fastener Type	Diameter (mm)	Lower Deviation (mm)	Nominal (mm)	Upper Deviation (mm)
MS 14218/ PE 26118	3,2 (-4)	0,76	0,69	0,66
	4,0 (-5)	0,96	0,89	0,86
	4,8 (-6)	1,19	1,12	1,09
	5,6 (-7)	1,42	1,35	1,32
MS 14219/ PE 26119	3,2 (-4)	1,11	1,04	1,01
	4,0 (-5)	1,42	1,35	1,32
	4,8 (-6)	1,80	1,73	1,70
	5,6 (-7)	2,03	1,96	1,93
100 ° Countersunk				
Fastener Type	Diameter	Lower Deviation	Nominal	Upper Deviation
Hi-Lok/Hi-Lite (1)/Lockbolt, Shear Head	4,0 (-5)	1,12	1,04	0,96
	4,8 (-6)	1,27	1,19	1,11
	6,4 (-8)	1,53	1,45	1,37
	7,9 (-10)	1,74	1,66	1,58
	9,5 (-12)	1,98	1,90	1,82

(1) Except HL23 and HST23

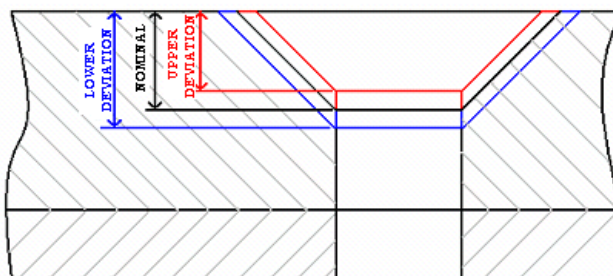


FIGURE 6 : FASTENER HEAD DEVIATIONS

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6.3.3 Fastener Head Depth Deviation Gauge

A depth gauge may be used to determine the relative rivet head position before bucking as regards the sheet surface at the countersunk edge, and for “Briles” rivets 1/8”, 5/32” and 3/16” nominal diameters, the measured values from 0,00 mm to - 0,04 mm are acceptable.

The Figure 7 shows the two steps to obtain the measured value. Initially the gauge shall be set to zero near the countersunk edge, followed by the positioning of the peak at the center of the non-bucking fastener head. The measured value shall be conforming to the values established in this paragraph.

NOTE

When using this method, the rivet that was used to calibrate the countersinking, shall be the same to be installed. This method shall not be used to control the riveting process.

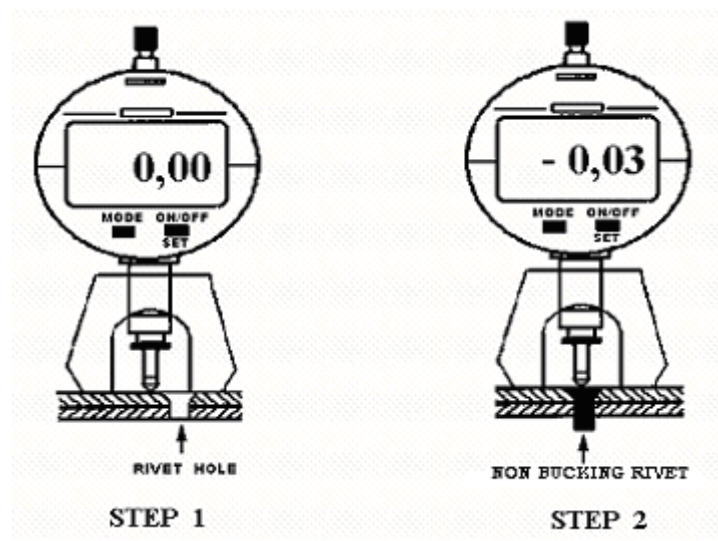


FIGURE 7 : FASTENER HEAD DEPTH DEVIATION GAUGE - MEASURE STEPS

7. VISUAL INSPECTION QUALIFICATION

The inspector and / or operator in charge of the inspection shall be approved in the Jaeger 1 or equivalent accuracy visual test, each two years at the minimum.

8. GENERAL NOTES

8.1 DRAWING CALLOUT

This Standard shall be called out on production drawings through:

NI 573 - Drill, countersink, and dimple for installation of rivets according to NE 03-006.

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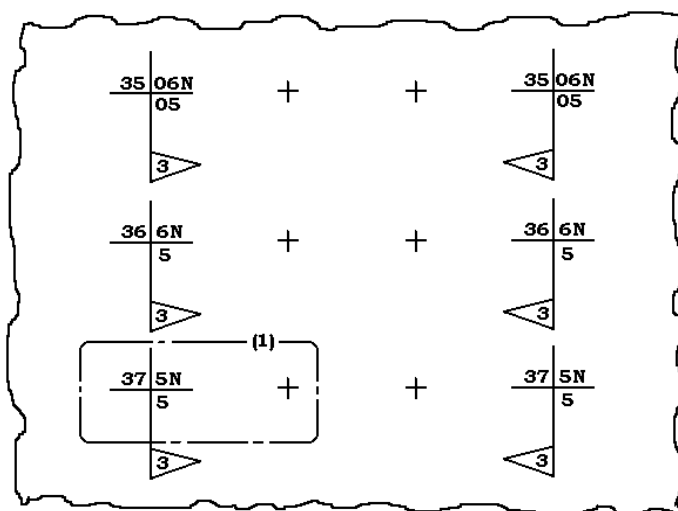
-ATTENTION-

For Hi-Lok, Hi-Lite, Lockbolt, and/or Hi-Tigue pins, besides the NI 573, it shall be indicated the dimensional limits and the surface roughness of the holes.

This information shall be provided in a table positioned on the upper right corner or above the title block, of the illustrative drawings. See example in Figure 8.

NOTES

- 1) The dash number relative to length of the rivet shall not be indicated in the Table. Indicate only the basic PN and the diameter dash number.
- 2) If the drawing contains rivets with the same basic PN and same diameter, installed in holes with different fit class, these rivets shall be evidenced on the drawing, except that rivet whose fit class is the majority, in the following way:
 - a) They shall be indicated by a dotted-and-dashed line; and
 - b) This line shall be interrupted by a numbered triangle for cross reference with the dimension table.



RIVET/PIN/BOLT			HOLE			
BASIC P/N + DIAMETER DASH NUMBER	DIAMETER (mm)		NI 573 FOR RIVETS AND PINS NI 720 FOR BOLTS			
			FIT CLASS	DIAMETER (mm)		Ra μm
	MIN	MAX		MIN	MAX	
LGPL2SP-V06-	4,788	4,813	II	4,813	4,839	1,6
HLT10-6-	4,940	4,966	-	4,864	4,890	1,2
HST10-5-	4,128	4,153	IV	4,102	4,128	1,6
HST10-5-	4,128	4,153	III	4,089	4,166	1,6

FIGURE 8 : EXAMPLE OF DIMENSIONAL LIMITS AND SURFACE ROUGHNESS INDICATION FOR DIAMETER HOLES

- 3) For drawings released before January, 6 of 1997 (Revision "K") without class fit or hole tolerance indication, class III shall be used.

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